PHASE 1:

- **What is a WLAN**: A Wireless Local Area Network is a technology that allows devices such as laptops, phones to connect to a network and the internet wirelessly, typically using radio waves via Wi-Fi standards within a limited area like a home or office. It uses access point to connect wireless cabled devices to wired network**.** WLAN has component such wireless controller, wireless adapter
  - **Technologies & Standards:** Primarily based on the IEEE 802.11 (Wi-Fi) standard, operating on 2.4 GHz, 5 GHz, and 6 GHz band AdHoc Mode: Devices connect directly to each other without a central router (peer-to-peer).
  - WLAN increases flexibility and ease of installation without physical cables but it offers lower security and are vulnerable to interference.

- **What is the difference between LAN and WLAN**:
  - A LAN (Local Area Network) connects devices within a small, defined area like an office or home using physical Ethernet cables, providing high speed and stability.
  - A WLAN (Wireless Local Area Network) is a type of LAN that uses radio waves (Wi-Fi) instead of cables for connectivity, offering greater mobility and flexibility.

<u>Key differences between LAN and WLAN</u>

1. **Connectivity:** LAN uses cables (Ethernet); WLAN uses radio waves.
2. **Mobility:** WLAN allows users to move freely; LAN requires devices to be secured.
3. **Installation:** LAN requires cabling infrastructure; WLAN is generally easier to set up, requiring only wireless access points.
4. **Security:** Wired LANs are generally more secure, while WLANs are more susceptible to unauthorized access as signals travel through the air.
5. **Performance:** LAN typically offers higher, more consistent speeds, whereas WLAN can be affected by interference, obstacles, and distance from the router.

- **What is a Wireless Access Point**: it is a networking hardware device that acts as a central hub, allowing Wi-Fi enabled devices (laptops, phones) to connect to a wired network using radio waves. It bridges wireless devices to a wired router or network switch, extending Wi-Fi coverage and capacity.
  Wireless Access point is Ideal for expanding Wi-Fi to "dead zones," supporting high-density environments with many users, or replacing multiple wired connections.
  - **Function:** It receives data from a wired cable (Ethernet) connected to a router and converts it into a wireless signal for devices, acting as a "bridge".
  - **Types**: Available as indoor (ceiling/wall mounted) or outdoor (waterproof) units.
- **What is SSID**: SSID (SERVICE SET IDENTIFIER) is the technical, user-friendly name of a Wi-Fi network that allows devices to identify and connect to it, SSID distinguishes one wireless network from another in a given area. It is 32-character maximum, and case sensitive(include letters ,numbers and special characters)
  - **Broadcasting & Visibility:** Routers usually broadcast their SSID so devices can find them, though this can be hidden for increased privacy.
  - **Connectivity:** Devices look for this name in Wi-Fi settings to connect, often requiring a password (pre-shared key) to gain access.
  - **Dual-Band & Multiple SSIDs:** A single router can have multiple SSIDs, typically one for 2.4GHz and one for 5GHz

- **What is WPA2/WPA3?**
  - ✓ WPA2(Wi-Fi Protected Acess2):It is a widely used ,secure wireless encryption protocol standard that secures network traffic through Advanced Encryption Standards(AES).it requires all certified Wi-Fi equipment to protect data from eavesdropping and unauthorized access.
  - ✓ WPA3(Wi-Fi Protected Acess3):It is the latest, most secure wireless security protocol introduced by the **Wi-Fi Alliance in 2018**,designed to replace the WPA2.it enhances network security through superior encryption ,Improves protection against brute force password guessing through Simultaneous Authentication of Equals (SAE)and better security for public networks using individualized encryption.

Differences between WPA2 and WPA3

- WPA3 offers enhanced security features, improved encryption methods, and better protection against modern threats compared to WPA2, making it the preferred choice for securing wireless networks.
- WPA2 uses AES encryption with a shared password, while WPA3 (released in 2018) provides superior security via stronger cryptographic algorithms, individualized data encryption (even on open networks), and better protection against offline password-guessing attacks.
- Security Strength: WPA2 is vulnerable to modern, offline attacks like KRACK, while WPA3 uses the Simultaneous Authentication of Equals (SAE) protocol to provide much stronger security.
- Encryption: WPA2 typically uses a 128-bit key. WPA3 increases this with 192-bit security for enterprise, and stronger encryption keys (128-bit to 256-bit), making them much harder to break.

- **What is difference between 2.4 GHz and 6GHz**

- Speed: 5 GHz is significantly faster than 2.4 GHz, making it better for high-bandwidth activities like 4K streaming, gaming, and video conferencing.
- Range: 2.4 GHz has a longer range and penetrates solid objects (walls, floors) better than 5 GHz, which is better for large homes.
- Interference: 2.4 GHz is more crowded, often facing interference from appliances like microwaves, Bluetooth devices. 5 GHz has more channels, resulting in less congestion